

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 30 Minutes
Total Marks:20

Annexure A

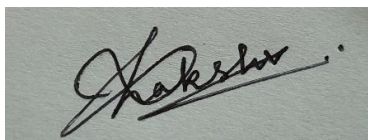
Descriptive Written Test

Code of Conduct:

I S. LAKSHITA(192524108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CO1 – ASSESSMENT TOOL 3

Descriptive Written Test

COURSE NAME: Artificial Intelligence

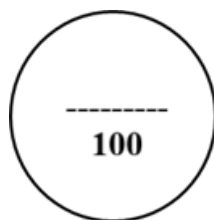
COURSE CODE: CSA17

QUESTIONS: 2

Duration: 45 Minutes

Total Marks: 20

Criteria	Weightage	Q1 (10)	Q2 (10)	Total Marks (20)
Understanding of Concepts	25% (2.5)	/2.5	/2.5	/5
Explanation	25% (2.5)	/2.5	/2.5	/5
Application	20% (2)	/2	/2	/4
Organization	15% (1.5)	/1.5	/1.5	/3
Accuracy	10% (1)	/1	/1	/2
Clarity	5% (0.5)	/0.5	/0.5	/1
Total per Question	10 Marks	/10	/10	/20



ANSWERS

1. Smart Home Automation System Using Intelligent Agents

Introduction

A smart home automation system uses Artificial Intelligence (AI) to automatically control lighting, temperature, and security based on user behavior and environmental conditions. The system receives real-time inputs from sensors such as motion detectors, temperature sensors, smoke detectors, and security cameras. It then analyzes this information and performs suitable actions to improve comfort, reduce energy consumption, and enhance home security. Different types of intelligent agents can be used to make decisions, each having its own strengths and limitations.

1. Simple Reflex Agent

A **Simple Reflex Agent** is the most basic type of intelligent agent that makes decisions based only on the current percept. It follows predefined **IF–THEN rules** and does not remember previous events or learn from experience. In a smart home, this agent can switch on lights when motion is detected, turn them off when no movement is sensed, activate the air conditioner when the temperature rises above a limit, or trigger an alarm if smoke is detected. It provides fast responses but cannot adapt to changing user preferences.

Advantages

- Responds quickly to real-time inputs.
- Simple and easy to implement.
- Requires less memory and computation.
- Suitable for routine automation tasks.

Limitations

- Cannot remember previous states.
- Does not learn user preferences.
- Performs poorly in dynamic environments.
- Cannot plan future actions.

2. Model-Based Reflex Agent

A **Model-Based Reflex Agent** improves upon the Simple Reflex Agent by maintaining an internal model of the environment. It stores previous information and updates it whenever new sensor data is received. In a smart home, it can remember whether lights are already on, monitor room occupancy, and track temperature history before making decisions. This allows the system to avoid unnecessary actions and operate more efficiently even when all information is not directly available.

Advantages

- Maintains memory of previous states.
- Works well in partially observable environments.
- Improves decision-making accuracy.
- Reduces unnecessary switching of devices.

Limitations

- More complex than a Simple Reflex Agent.
- Requires additional memory and processing.
- Does not independently optimize multiple objectives.

3. Goal-Based Agent

A **Goal-Based Agent** makes decisions by considering specific goals that need to be achieved. Instead of simply reacting to sensor inputs, it evaluates different possible actions and selects the one that best satisfies the desired objective. In a smart home, its goals may include maintaining a comfortable temperature, reducing electricity consumption, and ensuring home security. For example, when everyone leaves the house, the agent can lock doors, switch off unnecessary appliances, and activate the security system to achieve the goal of protecting the home.

Advantages

- Supports planning and intelligent decision-making.
- Achieves long-term goals efficiently.
- Flexible in changing environments.
- Suitable for complex automation systems.

Limitations

- Requires more computation than reflex agents.
- Decision-making is comparatively slower.
- Depends on planning and search techniques.

4. Utility-Based Agent

A **Utility-Based Agent** selects the action that provides the highest overall benefit by considering multiple factors such as comfort, energy efficiency, security, and cost. Instead of focusing on a single goal, it compares different alternatives and chooses the most beneficial one. In a smart home, it may decide to use ceiling fans instead of the air conditioner when the weather is pleasant, thereby saving electricity while maintaining user comfort. It can also learn user preferences over time and continuously improve its decisions.

Advantages

- Produces optimal decisions.
- Balances comfort, security, and energy efficiency.
- Learns and adapts to user preferences.
- Improves overall system performance.

Limitations

- More difficult to design and implement.
- Requires higher computational resources.
- Utility calculations can be complex.

Comparison of Intelligent Agents

Agent Type	Memory	Planning	Learning	Application in Smart Home
Simple Reflex Agent	No	No	No	Basic lighting and alarm control
Model-Based Reflex Agent	Yes	Limited	No	Occupancy monitoring and device management
Goal-Based Agent	Yes	Yes	Limited	Security planning and home automation

Utility-Based Agent	Yes	Yes	Yes	Comfort, energy optimization, and smart decision-making
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Hybrid Intelligent Agent Approach

A **Hybrid Intelligent Agent** combines the features of all four intelligent agents to improve the overall performance of the smart home system. The **Simple Reflex Agent** provides immediate responses during emergencies such as fire or gas leakage. The **Model-Based Reflex Agent** remembers previous environmental conditions and appliance status for better decision-making. The **Goal-Based Agent** plans actions to achieve objectives such as securing the house or reducing energy consumption. Finally, the **Utility-Based Agent** evaluates different alternatives and selects the option that provides the maximum benefit by balancing comfort, energy efficiency, and security. By combining these capabilities, the system becomes more intelligent, reliable, and adaptable.

Justification

A **Hybrid Intelligent Agent** is the most effective choice for a smart home automation system because no single agent type can efficiently handle all situations. The Simple Reflex Agent ensures quick responses, the Model-Based Reflex Agent improves decisions using memory, the Goal-Based Agent performs intelligent planning, and the Utility-Based Agent optimizes multiple objectives simultaneously. Together, these agents create a system that is capable of responding quickly, learning user behavior, reducing energy consumption, and maintaining a secure and comfortable home environment.

Conclusion

Smart home automation requires intelligent systems that can respond to changing environmental conditions while meeting user needs. Although each intelligent agent has its own strengths, a **Hybrid Intelligent Agent** offers the best solution by combining fast responses, memory, planning, and optimization. This approach improves user comfort, enhances home security, and ensures efficient energy management, making it the most suitable choice for modern smart home automation systems.

2.Robot Navigation in an Unknown Maze Using Uninformed Search Strategies

Introduction

A robot placed in an unknown maze must find a path to the exit without having prior knowledge of the maze layout. To solve this problem, the robot explores different paths using search algorithms. **Uninformed search strategies** do not have any information about the location of the goal; they rely only on the available states and search systematically until the exit is found. Two commonly used uninformed search algorithms are **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)**.

1. Uniform Cost Search (UCS)

Uniform Cost Search (UCS) is an uninformed search algorithm that expands the node with the **lowest cumulative path cost** from the start node. Instead of selecting the shortest path based on depth, UCS always chooses the path with the minimum total cost. It continues expanding

nodes until the goal is reached, ensuring that the least-cost solution is found when all path costs are non-negative.

In an unknown maze, the robot explores different routes while keeping track of the total movement cost of each path. If some paths contain obstacles or rough terrain with higher costs, UCS avoids them unless necessary and eventually finds the least expensive path to the exit.

Advantages

- Finds the optimal least-cost path.
- Complete algorithm if path costs are positive.
- Suitable for problems with varying movement costs.
- Guarantees the best solution.

Limitations

- Requires high memory to store nodes.
- Can be slow for large search spaces.
- Expands many nodes before reaching the goal.

2. Iterative Deepening Search (IDS)

Iterative Deepening Search (IDS) combines the advantages of **Depth-First Search (DFS)** and **Breadth-First Search (BFS)**. It repeatedly performs depth-limited searches by gradually increasing the depth limit until the goal is found. Since the robot does not know the location of the exit, IDS explores the maze level by level without using excessive memory.

In the maze, the robot first searches at depth 0, then depth 1, followed by depth 2, and so on until it reaches the exit. Although some nodes are explored multiple times, IDS is efficient in terms of memory and is useful when the goal depth is unknown.

Advantages

- Requires very little memory.
- Complete for finite search spaces.
- Useful when the goal depth is unknown.
- Simpler than many informed search methods.

Limitations

- Repeats exploration of upper-level nodes.
- Slower because of repeated searches.
- Does not consider path costs.
- May not find the least-cost path.

Advantages and Disadvantages in Terms of Time and Space Complexity

Uniform Cost Search (UCS)

Time Complexity: Depends on the branching factor and path costs, making it relatively high because many nodes may be expanded before reaching the goal.

Space Complexity: High, as all generated nodes must be stored in memory using a priority queue.

Summary

- Time Complexity: High
- Space Complexity: High
- Suitable for optimal path finding.

Iterative Deepening Search (IDS)

Time Complexity: Higher than DFS because upper-level nodes are explored repeatedly during each iteration.

Space Complexity: Low, since only the current search path is stored in memory.

Summary

- Time Complexity: Moderate to High
- Space Complexity: Low
- Suitable when available memory is limited.

Relation to Different Types of Intelligent Agents

Simple Reflex Agent

A Simple Reflex Agent makes decisions only from the current sensor input without remembering previous paths. In an unknown maze, it may repeatedly visit the same locations and become trapped in loops. Therefore, it is not suitable for maze navigation.

Model-Based Agent

A Model-Based Agent maintains an internal representation of the environment by remembering previously visited paths and obstacles. This allows the robot to avoid revisiting the same locations and improves navigation efficiency in an unknown maze.

Goal-Based Agent

A Goal-Based Agent focuses on achieving a specific objective, which is reaching the maze exit. It evaluates possible paths using search algorithms and selects actions that help achieve the goal efficiently. This makes it highly suitable for path planning problems.

Utility-Based Agent

A Utility-Based Agent considers multiple factors such as travel cost, time, energy consumption, and safety before choosing a path. It selects the route that provides the highest overall benefit rather than simply reaching the destination.

Effect of Agent Design on Decision-Making

The design of an intelligent agent has a significant impact on how the robot navigates the maze. A **Simple Reflex Agent** reacts quickly but cannot remember previous movements, resulting in inefficient navigation. A **Model-Based Agent** improves decision-making by storing information about explored paths. A **Goal-Based Agent** uses planning and search algorithms to systematically reach the exit, while a **Utility-Based Agent** evaluates multiple alternatives and selects the most beneficial route. Therefore, more advanced agent designs lead to smarter and more efficient decision-making.

Recommended Combination and Justification

The most effective combination for this problem is a **Goal-Based Agent with Uniform Cost Search (UCS)**.

A Goal-Based Agent continuously focuses on reaching the exit, while UCS guarantees the least-cost path even when different routes have different movement costs. This combination enables

the robot to make intelligent decisions, avoid unnecessary exploration, and reach the destination efficiently. If memory is limited and all movement costs are equal, a **Model-Based Agent with Iterative Deepening Search (IDS)** can also be an effective alternative because it requires less memory while still ensuring that the goal is eventually found.

Conclusion

Uninformed search strategies play an important role in solving navigation problems where no prior knowledge of the environment is available. **Uniform Cost Search** is best suited for finding the optimal least-cost path, whereas **Iterative Deepening Search** is more memory-efficient when the goal depth is unknown. Among the intelligent agents, a **Goal-Based Agent combined with Uniform Cost Search** provides the best balance of planning, accuracy, and optimal decision-making, making it the most suitable approach for robot navigation in an unknown maze.